Supplementary Information

Replacing beef: nutrition, emissions and the limits of a campus ban

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This document gives every calculation behind the main paper, step by step and with the numbers used. All numbers are produced by the code in the repository (`scripts/run_all.py`) and inserted into this document automatically, so the text, tables and figures cannot drift apart. Central values are reported first, followed by Monte Carlo medians and 90% intervals (5th–95th percentiles) where relevant. Units: kg CO_2e uses 100-year global warming potentials (GWP_{100}) unless stated; "g as eaten" means cooked weight for meat, fish and pulses.

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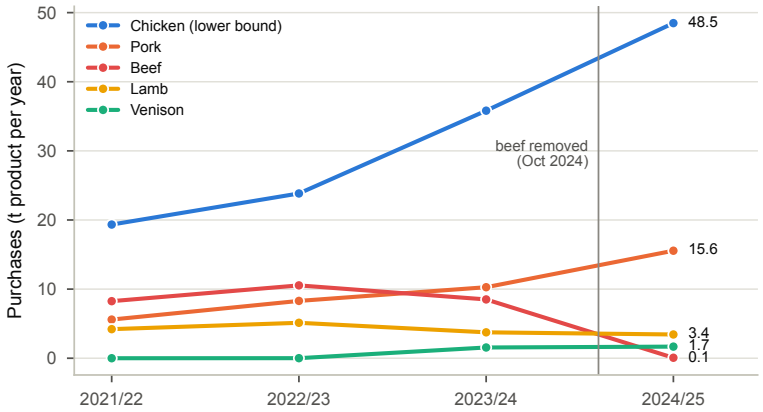
Supplementary Note 1. Imperial College London's beef policy and purchase data

Policy timeline. Imperial's Sustainable Food and Drink Policy (announced 7 December 2022) committed in-house catering to cut beef by 30% a year and to remove it by 2025–26[1]. The College's own analysis attributed 40% of catering CO_2e to fresh meat, which was only 8% of purchases, and 38% of meat CO_2e to beef[2]. Taste Imperial removed beef from all of its menus in October 2024[2]. Wild venison was introduced in some dishes, and the standard meat portion was cut from 120 g to 100 g[3]. The 2026–31 policy binds only in-house kitchens, outlets and events. Outsourced outlets are encouraged to stop selling beef by 2031[4]. Imperial College Union's 568 bar stopped serving beef in March 2026[5].

Purchase data. We parsed the line-item food purchase records released under the Freedom of Information Act (requests IMPFOI-24-576 and IMPFOI-25-493)[6] with `scripts/foi_meat_purchases.py`. Pack weights were read from free text and multiplied by order quantities. Lines without a parseable weight were excluded, so the totals are lower bounds. Supplementary Table 1 and Supplementary Fig. 1 show the results.

Supplementary Table 1 | Taste Imperial meat purchases (kg product weight per academic year). 2021/22 is a partial year. Beef was removed in October 2024 (academic year 2024/25). "Unparsed" is the number of meat purchase lines without a readable pack weight. The final columns give the purchases' emissions at this study's central emission factors (Supplementary Note 5).

Year	Beef (t)	Venison (t)	Lamb (t)	Pork (t)	Chicken (t)	Unparsed	Beef tCO_2e	All meat tCO_2e
2021/22	8.26	0.00	4.2	5.6	19.3	62	286	713
2022/23	10.54	0.01	5.1	8.3	23.8	50	366	907
2023/24	8.51	1.55	3.7	10.3	35.8	43	295	929
2024/25	0.07	1.68	3.4	15.6	48.5	47	2	814



Supplementary Fig. 1 | Meat purchases by Taste Imperial before and after beef removal. Lines show kg of product per academic year, from FOI line-item data. Chicken values are lower bounds. The vertical line marks the removal of beef in October 2024.

What the purchase data can and cannot show. Beef purchases fell from 10.5 t (2022/23) and 8.5 t (2023/24) to 0.07 t in 2024/25, while venison rose to 1.7 t and pork and chicken purchases kept growing (to 15.6 t and at least 48.5 t). At this study's central emission factors the emissions embodied in meat purchases fell from 929 t to 814 t CO_2e despite the growth in volume. Chicken purchases were already rising by roughly 12 t a year before beef was removed, so a simple before–after comparison would attribute an existing trend to the policy. Supplementary Note 12 sets out designs that handle this.

Supplementary Note 2. Food composition and bioavailability

Nutrient contents a_{ik} of food i and nutrient k are per 100 g as eaten (Supplementary Table 3). Beef, the reference food, is lean beef mince (about 10% fat), cooked. Four nutrients are adjusted for bioavailability:

$$\text{protein}_i^{q} = \text{protein}_i \times \min(\text{DIAAS}_i,\, 1), \tag{S1}$$

$$\text{Fe}_i^{abs} = \text{Fe}_i \left[h_i\, a_{\text{haem}} + (1 - h_i)\, a_{\text{non-haem}} \right], \tag{S2}$$

$$\text{Zn}_i^{abs} = \text{Zn}_i\, a_{\text{Zn}}(c_i), \qquad \text{B}_{12}{}_i^{abs} = \text{B}_{12,i}\, a_{\text{B12}}(c_i), \tag{S3}$$

where DIAAS is the digestible indispensable amino acid score[7], h_i the haem fraction of iron, and $a(\cdot)$ the fractional absorption for the food class c_i (Supplementary Table 2). Truncating DIAAS at 1 counts only protein that is not limited by its first-limiting indispensable amino acid, the standard convention for single foods. It ignores complementation between foods in a mixed basket, so it is conservative for plant baskets.

Supplementary Table 2 | Bioavailability parameters. Central values with the range used in sensitivity analyses.

Parameter	Central	Range	Note	Ref.
haem abs	0.25	0.15–0.35	Haem iron absorption 25% (EFSA 2015; FAO/WHO 10-40%)	[8]
nonhaem abs	0.10	0.05–0.15	Non-haem iron; whole mixed diets 14-18%, vegetarian 5-12%	[9]
zinc abs animal	0.35	0.30–0.40	Low-phytate diets: FAO/WHO 50/30/15% at phytate:zinc <5/5-15/>15; IZiNCG 26-34% mixed diets	[10]
zinc abs mixed	0.25	0.20–0.30	Moderate-phytate foods (tofu, fortified foods); mycoprotein ileal Zn absorption 24% (Edwards & Cummings 2010)	[11]
zinc abs phytate	0.18	0.12–0.25	Unrefined cereal / legume diets: IZiNCG 18% (men), 25% (women)	[12]
zinc abs supplement	0.40	0.30–0.50	Zinc salts taken with a meal	[11]
b12 abs meat	0.60	0.50–0.70	Mutton 56-89%, chicken 61-66%	[13]
b12 abs fish	0.42	0.35–0.50	Trout 42%	[13]
b12 abs egg	0.09	0.04–0.20	Egg B12 poorly absorbed (<9%)	[13]
b12 abs dairy	0.65	0.50–0.70	Milk B12	[13]
b12 abs synthetic	0.50	0.40–0.60	Cyanocobalamin in fortified foods/supplements at low doses	[13]

Worked example: 100 g of cooked beef. Protein: $28.4 \times \min(0.99, 1) = 28.1$ g. Iron: $3.08 \times [0.65 \times 0.25 + 0.35 \times 0.10] = 0.61$ mg absorbable. Zinc: $6.84 \times 0.35 = 2.39$ mg absorbable. Vitamin B_{12}: $2.80 \times 0.60 = 1.68$ µg absorbable. These are the targets b_k for the bioavailability-adjusted nutrients; selenium (21.6 µg) and the other nutrients enter unadjusted.

Supplementary Note 3. Single-food equivalents

The mass of food i that supplies as much of nutrient k as 100 g of beef is

$$g_{ik} = 100\,\text{g} \times \frac{b_k}{a_{ik}}, \qquad b_k = a_{\text{beef},k}, \tag{S4}$$

and the mass that matches *every* target nutrient is $g_i = \max_k g_{ik}$. The nutrient that attains the maximum is the binding nutrient. Supplementary Table 4 lists g_{ik} for all foods. "never" marks a food that contains none of the nutrient.

Worked example. Lentils contain $9.02 \times 0.58 = 5.23$ g quality-adjusted protein per 100 g, so matching beef's 28.1 g needs $100 \times 28.1/5.23 = 537$ g. Their 1.27 mg zinc is absorbed at 0.18 (0.229 mg), so matching beef's 2.39 mg absorbable zinc needs 1,047 g. Lentils contain no vitamin B_{12}, so no amount of lentils matches every nutrient ("never"). Chicken breast, by contrast, matches protein with 91 g but needs 684 g for zinc and 824 g for vitamin B_{12}.

Supplementary Note 4. Nutrient-matching linear programmes

For a set A of allowed foods, the basket $x_i \geq 0$ (units of 100 g as eaten, or tablets) that replaces 100 g of beef solves

$$\min_{x,\,s} \sum_{i \in A} c_i x_i + M \sum_k s_k \tag{S5}$$

$$\text{s.t.} \sum_{i \in A} \frac{a_{ik}}{b_k} x_i + s_k \geq 1 \quad \forall k \in K, \qquad 0 \leq s_k \leq 1, \tag{S6}$$

$$(1 - \delta_{\text{lo}})\, E_{\text{beef}} \leq \sum_i e_i x_i \leq (1 + \delta_{\text{hi}})\, E_{\text{beef}}, \tag{S7}$$

$$\sum_i \text{SFA}_i\, x_i \leq \rho\, \text{SFA}_{\text{beef}}, \qquad \sum_i \text{Na}_i\, x_i \leq \text{Na}_{\text{beef}} + \Delta_{\text{Na}}, \tag{S8}$$

$$0 \leq x_i \leq \bar{x}_i, \tag{S9}$$

Supplementary Table 3 | Composition of the reference beef and of the alternative foods, per 100 g as eaten (one tablet for supplements). Sources are given in the last column. "Diet" is the food group used to define scenarios.

Food	Diet	kcal	Prot	Fat	SFA	Fib	Fe	Zn	B12	Nia	B6	B2	Se	P	K	Na	Ca	DIAAS	Haem	Zn class	Source
Beef mince	meat	230	28.4	12.0	4.7	0.0	3.08	6.84	2.80	6.8	0.43	0.19	21.6	250	433	87	16	0.99	0.65	animal	FDC 171794
Chicken breast	meat	165	31.0	3.6	1.0	0.0	1.04	1.00	0.34	13.7	0.60	0.11	27.6	228	256	74	15	1.08	0.45	animal	FDC 171477
Pork loin	meat	209	28.6	9.6	3.5	0.0	1.09	2.53	0.73	5.9	0.55	0.33	35.1	249	425	58	18	1.17	0.55	animal	FDC 168233
Salmon (farmed)	fish	206	22.1	12.4	2.4	0.0	0.34	0.43	2.80	8.0	0.65	0.14	41.4	252	384	61	15	1.00	0.35	animal	FDC 175168
Egg	egg	155	12.6	10.6	3.3	0.0	1.19	1.05	1.11	0.1	0.12	0.51	30.8	172	126	124	50	1.13	0.00	animal	FDC 173424
Cheddar	dairy	403	22.9	33.3	18.9	0.0	0.14	3.64	1.10	0.1	0.07	0.43	28.5	455	76	653	710	1.20	0.00	animal	FDC 173414
Milk	dairy	50	3.3	2.0	1.3	0.0	0.02	0.48	0.53	0.1	0.04	0.18	2.5	92	140	47	120	1.23	0.00	animal	FDC 171267
Tofu	plant	144	17.3	8.7	1.3	2.3	2.66	1.57	0.00	0.4	0.09	0.10	17.4	190	237	14	683	0.97	0.00	mixed	FDC 172475
Lentils	plant	116	9.0	0.4	0.1	7.9	3.33	1.27	0.00	1.1	0.18	0.07	2.8	180	369	2	19	0.58	0.00	phytate	FDC 172421
Chickpeas	plant	164	8.9	2.6	0.3	7.6	2.89	1.53	0.00	0.5	0.14	0.06	3.7	168	291	7	49	0.76	0.00	phytate	FDC 173757
Kidney beans	plant	127	8.7	0.5	0.1	7.4	2.94	1.07	0.00	0.6	0.12	0.06	1.2	142	403	2	28	0.88	0.00	phytate	FDC 175194
Green peas	plant	78	5.2	0.3	0.0	4.5	1.52	0.67	0.00	1.5	0.11	0.10	1.0	77	110	72	24	0.68	0.00	phytate	FDC 170017
Mycoprotein	plant	98	14.5	1.7	0.4	7.2	0.50	7.00	0.00	0.0	0.00	0.00	18.0	0	0	50	0	0.99	0.00	mixed	Quorn labels; Finnigan 2019; Mayer Labba 2022
Pea-protein mince	plant	196	16.0	12.0	2.9	1.2	0.00	0.00	0.00	0.0	0.00	0.00	0.0	0	0	300	0	0.73	0.00	mixed	UK retailer labels
Peanuts	plant	587	24.4	49.7	7.7	8.4	1.58	2.77	0.00	14.4	0.47	0.20	9.3	363	634	6	58	0.43	0.00	phytate	FDC 173806
Pumpkin seeds	plant	559	30.2	49.0	8.7	6.0	8.82	7.81	0.00	5.0	0.14	0.15	9.4	1,230	809	7	46	0.60	0.00	phytate	FDC 170556
Oats	plant	379	13.2	6.5	1.1	10.1	4.25	3.64	0.00	1.1	0.10	0.15	28.9	410	362	6	52	0.57	0.00	phytate	FDC 173904
Fortified cereal	fortified	358	10.0	2.1	0.4	17.0	8.00	2.50	2.10	13.0	0.00	1.20	0.0	449	533	400	44	0.45	0.00	mixed	UK label (Kellogg's); FDC 173888
Soy drink (fortified)	fortified	42	3.3	1.9	0.3	0.5	0.42	0.26	0.38	0.4	0.03	0.21	2.3	43	122	36	120	0.95	0.00	mixed	UK label (Alpro); FDC 172456
Venison	meat	187	26.4	8.2	4.0	0.0	3.35	5.20	2.32	9.3	0.47	0.33	10.3	228	364	78	14	1.10	0.65	animal	FDC 172603
B12 tablet	supplement	0	0.0	0.0	0.0	0.0	0.00	0.00	10.00	0.0	0.00	0.00	0.0	0	0	0	0	0.00	0.00	animal	assumption
Zinc tablet	supplement	0	0.0	0.0	0.0	0.0	0.00	10.00	0.00	0.0	0.00	0.00	0.0	0	0	0	0	0.00	0.00	supplement	assumption

Supplementary Table 4 | Grams of each food (as eaten) that supply the nutrient content of 100 g cooked beef. Protein, iron, zinc and vitamin B_{12} are bioavailability-adjusted. The last two columns give the amount needed to match all ten nutrients and the binding nutrient. Values above 2,000 g are shown as ">2k".

Food	Prot.*	Fe*	Zn*	B12*	Nia	B6	B2	Se	P	K	All	Binding
Venison	106	92	132	121	73	91	58	210	110	119	210	Selenium
Pork loin	98	306	270	384	115	78	58	62	100	102	384	B_{12} (absorbable)
Chicken breast	91	349	684	824	50	71	168	78	110	169	824	B_{12} (absorbable)
Salmon (farmed)	127	1,173	1,591	143	85	66	141	52	99	113	1,591	Zinc (absorbable)
Soy drink (fortified)	897	1,448	>2k	884	1,600	1,381	91	939	581	355	>2k	Zinc (absorbable)
Egg	223	511	651	1,682	>2k	354	37	70	145	344	>2k	Niacin
Cheddar	123	>2k	188	235	>2k	648	45	76	55	570	>2k	Niacin
Milk	852	>2k	1,425	488	>2k	1,126	103	864	272	309	>2k	Iron (absorbable)
Tofu	168	229	610	never	1,785	465	187	124	132	183	never	B_{12} (absorbable)
Lentils	537	183	1,047	never	642	240	262	771	139	117	never	B_{12} (absorbable)
Chickpeas	418	210	869	never	1,293	308	303	584	149	149	never	B_{12} (absorbable)
Green peas	803	400	1,985	never	459	379	191	>2k	325	394	never	B_{12} (absorbable)
Mycoprotein	196	1,217	137	never	never	never	never	120	never	never	never	B_{12} (absorbable)
Pea-protein mince	241	never	never	never	never	never	never	never	never	never	never	Iron (absorbable)
Peanuts	268	385	480	never	47	92	97	232	69	68	never	B_{12} (absorbable)
Pumpkin seeds	155	69	170	never	136	299	125	230	20	54	never	B_{12} (absorbable)
Oats	374	143	365	never	607	428	123	75	61	120	never	B_{12} (absorbable)
Fortified cereal	625	76	383	160	52	never	16	never	56	81	never	Vitamin B6
Kidney beans	369	207	1,243	never	1,176	357	329	1,800	176	107	never	B_{12} (absorbable)

where K is the set of ten target nutrients (bioavailability-adjusted protein, iron, zinc and vitamin B_{12}, plus niacin, vitamin B_6, riboflavin, selenium, phosphorus and potassium), c_i the emissions per unit (minimum-GHG objective) or the mass per unit (minimum-mass objective), e_i energy, and $\bar{x}_i$ a portion cap. Settings: energy band $\delta_{\text{lo}} = 0.15$, $\delta_{\text{hi}} = 1.00$ (relaxed to 2.00 in sensitivity analysis); saturated fat ratio $\rho = 1.00$; extra sodium allowance $\Delta_{\text{Na}} = 150\,\text{mg}$; portion caps in Supplementary Table 3. The shortfall slacks s_k carry a penalty $M = 100\,\text{kg CO}_2\text{e}$ per unit of fractional shortfall. Any basket's emissions are below 1 kg CO_2e, so the slacks are used only when no feasible basket exists, and they then report the unavoidable shortfall. The programmes are solved with HiGHS through `scipy.optimize.linprog`.

Scenarios restrict A by food group: omnivore without ruminant meat (poultry, pork, fish, eggs, dairy, plants, fortified foods); poultry and pork only; pescatarian; vegetarian; vegan with fortified foods; vegan with fortified foods and supplements; and vegan from whole foods only. Supplementary Table 5 gives the solutions and Supplementary Table 6 the sensitivity analyses.

Supplementary Table 6 | Sensitivity of basket emissions (kg CO_2e per 100 g beef replaced) and mass (g) to bioavailability and energy assumptions. Infeasible baskets (shortfall in at least one nutrient) are marked with †.

Scenario	Central	No bioavail. adj.	Plant-unfavourable	Plant-favourable	Energy ±15%	Energy ≤3×	All ten nutrients
Poultry & pork only	4.02 / 256†	4.02 / 256†	4.02 / 256†	4.02 / 256†	2.16 / 127†	4.31 / 277†	4.02 / 256†
Omnivore, foods only	1.59 / 548†	1.43 / 439	3.00 / 360†	1.54 / 447	0.79 / 178†	2.25 / 654	1.64 / 446†
Vegetarian, foods only	1.46 / 646†	1.47 / 699†	1.45 / 576†	1.30 / 684†	0.60 / 438†	1.61 / 872†	1.13 / 455†
Vegan incl. fortified foods	0.69 / 448†	0.62 / 410†	0.55 / 344†	0.67 / 462†	0.60 / 438†	0.79 / 635†	0.55 / 421†
Vegan, whole foods only	0.38 / 292†	0.32 / 274†	0.54 / 422†	0.36 / 272†	0.46 / 236†	0.38 / 292†	0.59 / 395†
Poultry/pork-led	0.97 / 228	0.80 / 202	1.36 / 361†	0.96 / 175	1.13 / 199†	0.97 / 228	0.97 / 226
Fish-led	1.58 / 256	1.24 / 183	1.65 / 294†	1.56 / 224	1.05 / 189†	1.58 / 256	1.58 / 256
Dairy/egg-led	0.81 / 296	0.64 / 282	0.87 / 356†	0.81 / 296	0.47 / 208†	0.81 / 296	0.82 / 291†
Plant-based	0.36 / 290	0.32 / 274	0.51 / 373†	0.36 / 272	0.58 / 190†	0.36 / 290	0.59 / 409

Supplementary Note 5. Emission factors and the UK beef supply mix

Alternative foods. Emission factors (EFs) are kg CO_2e per kg as purchased, cradle to retail (Supplementary Table 7). The central values are UK- or Europe-relevant where available, with Poore & Nemecek's global means[14] as a comparison. Emissions per LP unit as eaten are

$$\text{GHG}_i = \frac{m_i}{y_i\,(1-w)}\,\text{EF}_i, \tag{S10}$$

with unit mass $m_i = 0.1\,\text{kg}$, cooking yield y_i (kg as eaten per kg purchased), and household waste fraction $w = 0.10$ applied equally to all foods.

Beef consumed in the UK. The EF of beef is built from the origin and production-system mix of UK beef consumption (Supplementary Table 8). Farm-gate footprints per kg carcass weight equivalent (cwe) are converted to retail boneless meat and given a post-farm-gate adder:

$$\text{EF}_j^{\text{retail}} = \frac{\text{EF}_j^{\text{farm}}}{r} + p, \qquad \text{EF}_{\text{beef}} = \sum_j s_j\,\text{EF}_j^{\text{retail}}, \tag{S11}$$

where $r = 0.70$ is the carcass-to-retail yield, $p = 3.0\,\text{kg CO}_2\text{e kg}^{-1}$ the post-farm-gate emissions, and s_j the market share of segment j. The *consequential* variant multiplies each farm-gate EF by an avoidable fraction f_j. Beef from the dairy herd is a co-product of milk. If its beef is no longer demanded, the cow's emissions are reallocated to milk rather than avoided, so only the calf-rearing and finishing emissions disappear.

Supplementary Table 5 | Optimised baskets replacing 100 g of cooked beef. Composition in g as eaten (supplements in tablets), emissions with central emission factors, and the supply of each target nutrient relative to beef (1.00 = equal).

Scenario	Composition per 100 g beef replaced	kg CO_2e	g	kcal ratio	Prot.*	Fe*	Zn*	B12*	Nia	B6	B2	Se	P	K	Key matched
Poultry & pork only	171 g chicken breast, 85 g pork loin	4.02	256	2.00	2.75	0.77	0.56	0.43	4.19	3.50	2.48	3.57	2.41	1.85	no
Omnivore, foods only	300 g milk, 140 g lentils, 85 g mycoprotein, 12 g salmon, 11 g fortified cereal	1.59	548	2.00	1.16	1.00	1.00	0.77	0.61	1.04	4.21	1.47	2.43	2.41	no
Vegetarian, foods only	300 g milk, 148 g lentils, 112 g soy drink, 84 g mycoprotein, 3 g fortified cereal	1.46	646	2.00	1.18	1.00	1.00	0.76	0.39	0.96	4.87	1.35	2.41	2.58	no
Vegan incl. fortified foods	300 g soy drink, 94 g mycoprotein, 29 g pumpkin seeds, 20 g fortified cereal, 5 g lentils	0.69	448	2.00	1.05	1.00	1.00	0.46	0.79	0.34	4.80	1.24	2.36	1.68	no
Vegan, whole foods only	116 g mycoprotein, 95 g kidney beans, 81 g lentils	0.38	292	1.43	1.00	1.00	1.00	0.00	0.21	0.61	0.60	1.12	1.13	1.58	no
Poultry/pork-led	118 g chickpeas, 54 g chicken breast, 51 g lentils, 4 g mycoprotein, 0.4 zinc tablet, 0.3 b12 tablet	0.97	228	1.51	1.00	1.00	1.00	1.00	1.27	1.36	0.91	1.00	1.66	1.55	yes
Fish-led	103 g lentils, 77 g kidney beans, 76 g salmon, 0.5 zinc tablet, 0.2 b12 tablet	1.58	256	1.63	1.00	1.00	1.00	1.00	1.13	1.80	1.17	1.64	1.95	2.27	yes
Dairy/egg-led	184 g kidney beans, 112 g egg, 0.4 zinc tablet, 0.3 b12 tablet	0.81	296	1.77	1.00	1.11	1.00	1.00	0.17	0.83	3.56	1.69	1.81	2.04	yes
Plant-based	126 g kidney beans, 98 g mycoprotein, 65 g chickpeas, 0.3 b12 tablet, 0.1 zinc tablet	0.36	290	1.58	1.00	1.00	1.00	1.00	0.16	0.57	0.60	1.00	1.15	1.61	yes

Supplementary Table 7 | Emission factors (kg CO_2e per kg as purchased). Central value, 90% interval used in Monte Carlo sampling (log-normal), Poore & Nemecek global mean, cooking yield and greenhouse-gas composition used for metric conversions.

Food	Central	Median	5th–95th	P&N mean	Yield	CO_2/CH_4/N_2O	GWP	Note
beef mince	34.70	34.70	24.00–52.00	66.39	0.69	0.20/0.58/0.22	AR5	UK supply mix (overridden in model); ef_pn = 50:50 beef/dairy herd
chicken breast	9.87	7.52	3.95–20.82	9.87	0.73	0.61/0.06/0.33	AR5cc	Poultry meat
pork loin	12.31	10.57	6.91–23.79	12.31	0.80	0.49/0.25/0.26	AR5cc	Pig meat
salmon	13.63	7.89	5.41–32.64	13.63	0.80	0.70/0.10/0.20	AR5cc	Fish (farmed)
egg	4.67	4.21	2.85–8.49	4.67	0.88	0.55/0.10/0.35	AR5cc	Eggs; cook_yield = edible share of shell egg
cheddar	23.88	18.64	10.21–58.79	23.88	1.00	0.30/0.50/0.20	AR5cc	Cheese
milk	3.15	2.65	1.51–7.00	3.15	1.00	0.30/0.50/0.20	AR5cc	Milk
tofu	3.16	2.58	1.41–7.27	3.16	1.00	0.65/0.05/0.30	AR5cc	Tofu
lentils	1.79	1.39	0.89–4.00	1.79	2.73	0.65/0.05/0.30	AR5cc	Other pulses (dry weight)
chickpeas	1.79	1.39	0.89–4.00	1.79	2.31	0.65/0.05/0.30	AR5cc	Other pulses (dry weight)
kidney beans	1.79	1.39	0.89–4.00	1.79	2.50	0.65/0.05/0.30	AR5cc	Other pulses (dry weight)
peas	0.98	0.80	0.51–1.87	0.98	0.89	0.65/0.05/0.30	AR5cc	Peas
mycoprotein	1.89	1.89	1.26–2.60	–	1.00	0.80/0.05/0.15	AR5	Quorn Mince farm-to-shop (Carbon Trust / ClimatePartner verified)
pea mince	2.00	2.00	0.90–4.00	–	0.90	0.75/0.05/0.20	AR5	Plant-based mince; 0.75-0.98 factory gate (GFI) to 3.2-3.5 (Beyond/Impossible)
peanuts	3.23	3.29	1.42–6.15	3.23	1.00	0.65/0.05/0.30	AR5cc	Groundnuts
pumpkin seeds	3.23	3.29	1.42–6.15	–	1.00	0.65/0.05/0.30	AR5cc	Proxy: groundnuts (no seed LCA in P&N)
oats	2.48	2.59	0.80–4.30	2.48	1.00	0.65/0.05/0.30	AR5cc	Oatmeal (dry weight)
fortified cereal	2.36	1.91	1.07–4.61	–	1.00	0.70/0.05/0.25	AR5cc	Proxy: P&N wheat (bread) x 1.5 for processing and packaging
soy drink	0.98	0.91	0.51–1.74	0.98	1.00	0.65/0.05/0.30	AR5cc	Soymilk (per kg per L)
venison	3.00	3.00	1.00–12.00	–	0.83	1.00/0.00/0.00	AR5	Wild-shot deer: processing and transport only (assumption; Imperial cites <1/3 of beef)
lamb	39.72	40.61	23.70–60.16	39.72	0.72	0.20/0.55/0.25	AR5cc	Lamb & mutton (EDM cross-effects only)
b12 supplement	50.00	50.00	20.00–100.00	–	1.00	1.00/0.00/0.00	AR5	Per kg of tablets (one tablet 0.1 g)
zinc supplement	50.00	50.00	20.00–100.00	–	1.00	1.00/0.00/0.00	AR5	Per kg of tablets (one tablet 0.1 g)

Supplementary Table 8 | UK beef supply mix. Shares of consumption, farm-gate footprints and the resulting retail EFs.

Segment	Share	Farm EF (cwe)	90% range	Retail EF	Avoidable	Retail EF (conseq.)	Ref.
UK suckler herd	0.360	28.0	20–38	43.0	1.00	43.0	[15,16]
UK dairy-origin	0.360	16.0	10–24	25.9	0.70	19.0	[15,16]
Ireland	0.216	20.0	14–30	31.6	0.85	27.3	[17,18]
Other Europe	0.035	22.0	14–35	34.4	0.75	26.6	[17]
Australia & New Zealand	0.015	25.0	15–45	38.7	1.00	38.7	[17]
South America	0.015	60.0	30–150	88.7	1.00	88.7	[14,17]
UK consumption mix	1.001			**34.7**		**31.0**	

Worked calculation. For the UK suckler herd: $28.0/0.70 + 3.0 = 43.0$ kg CO_2e per kg retail beef. Weighting all six segments by their shares gives $EF_{beef} = 34.7$ kg CO_2e kg^{-1} (attributional) and 31.0 kg CO_2e kg^{-1} (consequential). For comparison, Poore & Nemecek's global means are 99.48 (beef herd) and 33.30 (dairy herd), 66.4 for a 50:50 mix, and their UK producers average 48.4 and 25.9 (mix 37.1), close to our bottom-up estimate. Per 100 g cooked beef: $34.7 \times 0.1/(0.69 \times 0.90) = 5.59$ kg CO_2e.

Supplementary Note 6. From one portion to London and the UK

Per-capita beef flows follow the supply chain from the national supply balance:

$$B_{\text{retail}} = B_{\text{cwe}}\, r, \qquad B_{\text{eaten}} = B_{\text{retail}}\,(1 - w)\, y_{\text{beef}}, \tag{S12}$$

where B_{cwe} is beef consumption in carcass-weight equivalent per person per year, r the retail yield, w household waste and y_{beef} the cooking yield. The annual change in emissions per person when beef is replaced by basket b is

$$\Delta e = \underbrace{10\, B_{\text{eaten}}\ \text{GHG}_b}_{\text{replacement}} - \underbrace{B_{\text{retail}}\ \text{EF}_{\text{beef}}}_{\text{beef}}, \tag{S13}$$

with GHG_b in kg CO_2e per 100 g of beef replaced (hence the factor 10 per kg). London and UK totals multiply by population.

Worked calculation (central values, omnivore basket). Beef supply is $B_{\text{cwe}} = 15.8$ kg per person per year, i.e. $B_{\text{retail}} = 15.8 \times 0.70 = 11.06$ kg bought and $B_{\text{eaten}} = 11.06 \times 0.90 \times 0.69 = 6.87$ kg cooked beef eaten (19 g a day). Beef emissions per person are $11.06 \times 34.7 = 384$ kg CO_2e a year. The poultry/pork-led basket emits 0.969 kg CO_2e per 100 g beef replaced, so the replacement emits $10 \times 6.87 \times 0.969 = 67$ kg. The change is =317 kg per person (−83% of beef's footprint). Multiplying by 9,089,736 Londoners gives =2.88 Mt CO_2e a year, i.e. 10.1% of London's territorial emissions (28.61 Mt) and 3.3% of its consumption-based emissions (88.0 Mt).

Supplementary Table 9 | Annual emissions change from replacing all beef eaten by Londoners with nutritionally equivalent baskets. Central values. Percentages are relative to London's territorial emissions (LEGGI 2023) and to London's consumption-based emissions.

Beef EF	Basket	Beef kg/p	Alt. kg/p	% vs beef	London beef Mt	Δ London Mt	% LEGGI	% CBE	Δ UK Mt	% UK terr.
UK mix	Poultry/pork-led	384	67	−83	3.49	−2.88	−10.1	−3.3	−22.0	−5.9
UK mix	Fish-led	384	109	−72	3.49	−2.50	−8.7	−2.8	−19.0	−5.1
UK mix	Dairy/egg-led	384	56	−86	3.49	−2.98	−10.4	−3.4	−22.7	−6.1
UK mix	Plant-based	384	25	−93	3.49	−3.26	−11.4	−3.7	−24.8	−6.7
UK mix, conseq.	Poultry/pork-led	343	67	−81	3.12	−2.51	−8.8	−2.9	−19.2	−5.1
UK mix, conseq.	Fish-led	343	109	−68	3.12	−2.13	−7.4	−2.4	−16.2	−4.3
UK mix, conseq.	Dairy/egg-led	343	56	−84	3.12	−2.61	−9.1	−3.0	−19.9	−5.3
UK mix, conseq.	Plant-based	343	25	−93	3.12	−2.89	−10.1	−3.3	−22.0	−5.9
Global (P&N) mix	Poultry/pork-led	734	67	−91	6.67	−6.07	−21.2	−6.9	−46.3	−12.4
Global (P&N) mix	Fish-led	734	109	−85	6.67	−5.69	−19.9	−6.5	−43.3	−11.6
Global (P&N) mix	Dairy/egg-led	734	56	−92	6.67	−6.17	−21.6	−7.0	−47.0	−12.6
Global (P&N) mix	Plant-based	734	25	−97	6.67	−6.45	−22.5	−7.3	−49.1	−13.2
Global beef herd	Poultry/pork-led	1,100	67	−94	10.00	−9.40	−32.8	−10.7	−71.6	−19.2
Global beef herd	Fish-led	1,100	109	−90	10.00	−9.01	−31.5	−10.2	−68.7	−18.4
Global beef herd	Dairy/egg-led	1,100	56	−95	10.00	−9.50	−33.2	−10.8	−72.4	−19.4
Global beef herd	Plant-based	1,100	25	−98	10.00	−9.77	−34.2	−11.1	−74.5	−19.9

Supplementary Note 7. Monte Carlo uncertainty and sensitivity

We drew 20,000 joint samples of all uncertain inputs: per-capita consumption, yields, waste, segment-level beef EFs, segment shares (Dirichlet), avoidable fractions, post-farm emissions, and the EF of every replacement food (log-normal, independent). Basket compositions are held at their central LP solutions. Supplementary Table 10 summarises the distributions, and Extended Data Fig. 2 of the main paper shows one-at-a-time sensitivities.

Supplementary Table 10 | Monte Carlo results for London (Mt CO_2e yr^{-1}). Median and 90% interval.

Basket	Beef accounting	Median	90% interval
Poultry/pork-led	attributional	−2.83	−3.72 to −2.05
Poultry/pork-led	consequential	−2.40	−3.24 to −1.68
Fish-led	attributional	−2.70	−3.63 to −1.84
Fish-led	consequential	−2.29	−3.14 to −1.43
Dairy/egg-led	attributional	−2.86	−3.74 to −2.16
Dairy/egg-led	consequential	−2.44	−3.24 to −1.79
Plant-based	attributional	−3.10	−3.96 to −2.40
Plant-based	consequential	−2.68	−3.46 to −2.04
Beef itself	attributional	3.30	2.58 to 4.17

Supplementary Note 8. Emission metrics: GWP_{100}, GWP_{20} and GWP*

Emission factors are reported in different GWP_{100} vintages. Poore & Nemecek used AR5 values with climate–carbon feedbacks (CH_4 34, N_2O 298). We convert each EF to a target metric with its gas composition (f_{CO_2}, f_{CH_4}, f_{N_2O}, shares of the source-metric total):

$$\mathrm{EF}^{\mathrm{target}} = \mathrm{EF}\left[f_{\mathrm{CO_2}} + f_{\mathrm{CH_4}}\frac{\mathrm{GWP}^{\mathrm{target}}_{\mathrm{CH_4}}}{\mathrm{GWP}^{\mathrm{source}}_{\mathrm{CH_4}}} + f_{\mathrm{N_2O}}\frac{\mathrm{GWP}^{\mathrm{target}}_{\mathrm{N_2O}}}{\mathrm{GWP}^{\mathrm{source}}_{\mathrm{N_2O}}}\right], \tag{S14}$$

using AR6 values for non-fossil CH_4 (GWP_{100} 27.0, GWP_{20} 79.7) and N_2O (273)[19].

GWP*. Methane is short-lived, so a sustained change in its emission *rate* has a different effect on warming from a pulse of CO_2. The warming-equivalent emissions of methane are[20,21]

$$E^*_t = \mathrm{GWP}_{100}\,(4.53\,E_t - 4.25\,E_{t-20})\,. \tag{S15}$$

If a constant methane source E_0 stops at $t = 0$, the change relative to it continuing is $-4.53\,\mathrm{GWP}_{100}E_0$ a year for 20 years and $-0.28\,\mathrm{GWP}_{100}E_0$ a year thereafter. Cumulatively this is about 4.5 times the GWP_{100} accounting over the first 20 years, falling to about 1.1 times after 100 years. Long-lived gases (CO_2, N_2O) are counted at GWP_{100}.

Worked calculation (London, UK supply mix). London's beef emits 3.49 Mt CO_2e a year, of which 55% is methane, 21% nitrous oxide and 24% CO_2 (UK inventory metric, CH_4=28). The methane is 3,487 kt × 0.545/28 = 67.9 kt CH_4 a year. Under AR6 GWP_{20} beef's footprint is 2.01 times its GWP_{100} value and London's saving becomes 6.4 Mt a year. Stopping the methane abruptly changes warming-equivalent emissions by $-4.53 \times 27 \times 67.9$ = =8.30 Mt a year for 20 years. Including long-lived gases and the replacement food, the cumulative reduction after 20 years is 186 Mt CO_2-we under GWP* versus 57 Mt under GWP_{100} (ratio 3.3); after 100 years the totals are 308 and 284 Mt.

Supplementary Note 9. Market response: equilibrium displacement model

Model. Each food market i has constant-elasticity supply $\mathrm{d}\ln Q^s_i = \eta_i\,\mathrm{d}\ln P_i$ and demand $\mathrm{d}\ln Q^d_i = \sum_j \varepsilon_{ij}\,\mathrm{d}\ln P_j + \Delta D_i/Q_i$, where ΔD_i is the exogenous shift in quantity demanded at unchanged prices (the campus shock) and ε_{ij} are Marshallian own- and cross-price elasticities[22,23]. Market clearing gives

$$(\mathrm{diag}(\eta) - \mathcal{E})\,\mathrm{d}\ln P = \Delta D/Q, \qquad \Delta Q = Q \circ \eta \circ \mathrm{d}\ln P. \tag{S16}$$

For one market without cross effects the change in production is $\Delta Q = \phi\,\Delta D$ with pass-through

$$\phi = \frac{\eta}{\eta - \varepsilon} \quad (\varepsilon < 0), \tag{S17}$$

and $1-\phi$ of the campus's forgone demand is consumed by other buyers because the price falls (market leakage). Perfectly elastic supply gives $\phi = 1$; fixed supply gives $\phi = 0$.

Shock. With G kg of beef removed from campus menus, a share s_{off} of forgone beef meals bought as beef off campus, and replacement dishes purchased per forgone meal according to the destination shares in Supplementary Table 15,

$$\Delta D_{\text{beef}} = -(1 - s_{\text{off}})\, G, \qquad \Delta D_i = N_{\text{beef}} \sum_d \pi_d\, q_{di} \quad (i \neq \text{beef}), \tag{S18}$$

where $N_{\text{beef}} = G/g_{\text{meal}}$ is the number of beef meals, π_d the destination shares and q_{di} the purchased kg of food i in a dish of type d. The net emissions change is $\sum_i \Delta Q_i\, \text{EF}_i$, which we decompose as

$$\Delta E = \underbrace{-G\,\text{EF}_{\text{b}}}_{\text{gross}} + \underbrace{\textstyle\sum_{i\neq b} \Delta D_i \text{EF}_i}_{\text{replacement food}} + \underbrace{s_{\text{off}} G\,\text{EF}_{\text{b}}}_{\text{off campus}} + \underbrace{(\Delta Q_{\text{b}} - \Delta D_{\text{b}})\text{EF}_{\text{b}}}_{\text{market leakage}} + \underbrace{\textstyle\sum_{i\neq b} (\Delta Q_i - \Delta D_i)\text{EF}_i}_{\text{replacement market adj.}}. \tag{S19}$$

Supplementary Table 11 | Market parameters. Baseline UK consumption, supply elasticities (short and long run) and own-price demand elasticities. Triangular distributions on the stated ranges.

Market	Q (kt)	η short run	η long run	ε own	ϕ SR	ϕ LR	Ref.
beef	766	0.25 (0.10–0.40)	2.0 (0.6–4.6)	−0.59 (−1.00 to −0.45)	0.30	0.77	24–26
lamb	180	0.20 (0.10–0.40)	1.5 (0.5–3.0)	−0.80 (−1.20 to −0.50)	0.20	0.65	27
pork	1,000	0.40 (0.15–0.75)	3.0 (1.5–4.0)	−0.78 (−1.10 to −0.55)	0.34	0.79	24–26
poultry	1,500	0.20 (0.06–0.40)	5.0 (2.0–13.1)	−0.68 (−0.95 to −0.45)	0.23	0.88	25,26,28
fish	500	0.30 (0.10–0.60)	2.0 (0.8–4.0)	−0.60 (−0.90 to −0.40)	0.33	0.77	28
eggs	780	0.30 (0.10–0.50)	3.0 (1.0–6.0)	−0.27 (−0.50 to −0.15)	0.53	0.92	28,29
dairy	6,000	0.20 (0.10–0.30)	1.0 (0.5–2.0)	−0.50 (−0.80 to −0.30)	0.29	0.67	28–30
plant	3,000	1.00 (0.50–2.00)	5.0 (2.0–10.0)	−0.60 (−0.90 to −0.30)	0.62	0.89	
venison	5	0.50 (0.30–1.00)	2.0 (1.0–4.0)	−0.80 (−1.20 to −0.50)	0.38	0.71	

Supplementary Table 12 | Cross-price demand elasticities ε_{ij} (row: quantity; column: price).

Quantity \ price	beef	lamb	pork	poultry
beef	0.000	−0.012	−0.049	−0.084
lamb	−0.009	0.000	0.000	0.000
pork	−0.098	0.000	0.000	0.000
poultry	−0.059	0.000	0.000	0.000

Worked calculation (central values). Taste Imperial bought $G = 9.5$ t of beef a year, i.e. $N_{\text{beef}} = 79{,}167$ meals of 0 g. With 7% of forgone beef meals bought as beef off campus, campus beef demand falls by 8.84 t. The long-run beef pass-through is $\phi = 2.0/(2.0 + 0.594) = 0.77$ without cross effects (0.77 with them), so production falls by 6.84 t. At the consequential EF of 31.0 kg CO_2e kg^{-1}, the components are: beef removed =295 t; replacement food +76 t; off-campus beef +21 t; beef market leakage +62 t; replacement market adjustment =7 t; net =143 t CO_2e a year. A procurement inventory would report =224 t.

Supplementary Table 13 | Campus beef removal: decomposition of the annual emissions change (t CO_2e yr^{-1}). Monte Carlo median and 90% interval (4,000 draws).

Component	Long run	Short run	Long run, no cross-price
Beef removed from menus	−293 (−373 to −233)	−294 (−374 to −233)	−293 (−373 to −236)
Replacement food bought	65 (42 to 103)	64 (41 to 103)	64 (43 to 104)
Beef bought off campus	17 (3 to 51)	18 (3 to 51)	18 (3 to 51)
Beef market leakage	61 (35 to 109)	197 (148 to 261)	61 (36 to 111)
Replacement market adjustment	−6 (−14 to −0)	−36 (−63 to −21)	−12 (−20 to −7)
Net change	−150 (−223 to −83)	−47 (−88 to −15)	−154 (−224 to −94)
Procurement accounting	−232 (−314 to −165)	−233 (−316 to −166)	−232 (−312 to −168)
Beef pass-through ϕ	0.78 (0.62 to 0.87)	0.28 (0.18 to 0.38)	0.78 (0.62 to 0.86)

Supplementary Note 10. Welfare of diners

Model. A diner choosing a dish on a meal occasion has utility $U_j = V_j - \alpha p_j + \epsilon_j$ with generalised-extreme-value errors that form a "meat" nest (beef and other meat dishes) with dissimilarity parameter $\lambda \in (0, 1]$ and other dishes as their own nests. The expected maximum utility is the log-sum, so removing the beef alternative(s) with prices held fixed gives the compensating variation per meal occasion[31,32]

$$\mathrm{CV} = \frac{1}{\alpha} \ln\Big(1 - P_M + P_M\big(1 - P_{B|M}\big)^{\lambda}\Big), \tag{S20}$$

where P_M is the meat-nest share and $P_{B|M} = P_B/P_M$ the beef share within the nest. For $\lambda = 1$ this reduces to the multinomial-logit result $\mathrm{CV} = \ln(1 - P_B)/\alpha$. As $\lambda \to 0$, other meat is a perfect substitute and $\mathrm{CV} \to 0$. The loss per forgone beef meal is $-\mathrm{CV}/P_B$.

Calibration. The marginal utility of money α comes from an assumed own-price elasticity of a beef dish,

$$\varepsilon_B = -\alpha\, p_B \left[\frac{1}{\lambda} - \Big(\frac{1}{\lambda} - 1\Big) P_{B|M} - P_B\right]. \tag{S21}$$

A diner can always walk off campus and buy a beef meal, so the loss per forgone beef meal cannot exceed the cost of that outside option (time plus any price premium). We report the logit estimate and the estimate capped at this cost.

Worked calculation (central values). With 1,000,000 main meals a year and 79,167 beef meals, $P_B = 0.079$; with $P_M = 0.55$, $\lambda = 0.6$, $p_B = 6.00$ and $\varepsilon_B = =1.5$, the calibration gives $\alpha = 0.168$ per £. Then $\mathrm{CV} = \ln(1 - 0.55 + 0.55(1 - 0.144)^{0.6})/0.168 = =0.300$ per meal occasion, or £3.78 per forgone beef meal. Because α is calibrated to the beef dish's own elasticity, the loss per forgone beef meal is close to $p_B/|\varepsilon_B| = 4.00$ for any λ (Supplementary Note 10). Capped at the outside option (£3.00), the annual loss is £237,500.

Supplementary Table 14 | Welfare and monetised climate effects of campus beef removal (Monte Carlo).

Quantity	Median	90% interval
Loss per forgone beef meal, nested logit (£)	3.25	2.09 to 5.35
Loss per forgone beef meal, capped (£)	2.75	1.92 to 3.92
Annual welfare loss, nested logit (£)	253,969	160,709 to 423,618
Annual welfare loss, capped (£)	214,059	144,507 to 314,739
Emissions avoided, long run (t CO_2e)	150	83 to 223
Monetised climate benefit (£)	40,818	20,608 to 68,436
Welfare cost per tonne avoided (£/t)	1,730	911 to 3,725
Welfare cost per tonne, capped (£/t)	1,444	818 to 2,896
P(capped loss > climate benefit)	100%	

Supplementary Note 11. Nutrient intake of campus diners

Supplementary Table 15 defines the protein component of each dish type. Accompaniments (starch, vegetables) are assumed unchanged. Nutrients per dish are $\sum_i q_{di}\, y_i\, a_{ik} \times 10$ (kg purchased × cooking yield × nutrient per 100 g). The expected change per forgone beef meal is $\sum_d \pi_d (n_d - n_{\text{beef}})$, with off-campus non-beef meals taking the average on-campus alternative and off-campus beef meals unchanged.

Supplementary Table 15 | Dish templates (kg purchased per serving) and destination shares π_d of forgone beef meals.

Dish	Protein component (purchased weight)	Share π_d
beef	120 g beef mince	–
fish	120 g salmon	0.06
other meat	72 g chicken breast, 24 g pork loin, 24 g venison	0.50
vegan	30 g kidney beans, 20 g lentils, 30 g pea-protein mince, 60 g tofu	0.10
vegetarian	35 g cheddar, 30 g chickpeas, 30 g egg, 40 g mycoprotein	0.20
off campus, beef	as beef	0.07
off campus, other	average of on-campus alternatives	0.07

Supplementary Table 16 | Nutrients per serving of each dish's protein component, and the change per forgone beef meal.

Dish	Energy (kcal)	Protein (g)	Protein, DIAAS-adj. (g)	Iron (mg)	Iron, absorbable (mg)	Zinc (mg)	Zinc, absorbable (mg)	Vitamin B12 (µg)	Saturated fat (g)	Fibre (g)	Sodium (mg)
beef	190.4	23.5	23.3	2.6	0.5	5.7	2.0	2.3	3.9	0.0	72.0
fish	197.8	21.2	21.2	0.3	0.0	0.4	0.1	2.7	2.3	0.0	58.6
other meat	164.1	27.0	27.0	1.4	0.3	2.0	0.7	0.8	2.0	0.0	65.6
vegan	297.9	26.1	21.8	5.6	0.6	2.4	0.5	0.0	1.6	11.6	92.0
vegetarian	334.8	23.3	21.7	2.6	0.3	5.4	1.4	0.7	7.8	8.1	286.1
Δ per forgone beef meal	29.08	1.99	1.41	−0.42	−0.21	−2.70	−1.08	−1.41	−0.53	3.01	44.09

Supplementary Note 12. Measuring what the policy did: an evaluation design

Nine outcomes, how to measure them and how to identify the policy's effect are listed in Supplementary Table 17. Three points matter most for Imperial.

1. **Counterfactual trends.** Meat purchases were rising before the ban (Supplementary Table 1), so effects must be estimated against pre-trends (interrupted time series) or comparison outlets that kept beef (outsourced outlets until 2031), not as before–after differences. The simultaneous cut in meat portions from 120 to 100 g is a confounder that needs its own term.
2. **Leakage cannot be measured on campus.** Off-campus purchases need panel data (consented card or diary data). Market leakage needs the equilibrium model of Supplementary Note 9, because the campus is far too small to move prices measurably.
3. **Power.** Detecting a change of 0.5 mg/day in zinc intake (SD 3.5 mg/day) with 80% power at $\alpha = 0.05$ needs $n = 2(z_{0.975} + z_{0.80})^2\sigma^2/\delta^2 = 770$ participants per group. This is feasible only if recruitment targets frequent campus diners, because the average change across all students is an order of magnitude smaller.

Supplementary Table 17 | Evaluation framework for the campus beef policy.

Outcome	Metric	Data	Identification
Campus purchases and emissions	kg purchased by product; t CO_2e (attributional)	Procurement / FOI line items	Interrupted time series with pre-trends; DiD vs. outsourced outlets
Dish choice and substitution	Shares of meat, fish, vegetarian, vegan dishes	Till data with anonymised card IDs	Event study around Oct 2024; nested-logit demand estimation
Off-campus leakage	Campus meals per regular diner; beef meals bought elsewhere	Card panel; consented spending diaries; short surveys	Panel DiD; bounds from footfall changes
Market leakage	Pass-through ϕ	Published elasticities; trade data	Equilibrium displacement model; consequential LCA
Nutrient intake	Protein, iron, zinc, B_{12}, fibre, saturated fat	Repeated 24-h recalls (Intake24)	Pre/post with comparison university; usual-intake modelling
Nutrient status	Ferritin, haemoglobin, holotranscobalamin, plasma zinc	Biomarker subsample	Pre/post in frequent beef eaters vs. others
Satisfaction	Meal satisfaction, perceived choice, complaints	Validated survey scales; free text	Repeated cross-sections; DiD
Welfare	Compensating variation (£ per meal)	Revealed choices; discrete-choice experiment	Logit log-sum; stated willingness to pay to restore beef
Equity	Effects by sex, culture, income, sport	Linked survey and till data	Heterogeneous treatment effects

Supplementary Note 13. Complete parameter list

Parameter	Value	Range	Unit	Note	Ref.
population.london	9,089,736		persons	ONS mid-2024 estimate, Greater London	33
population.uk	69,281,437		persons	ONS mid-2024 estimate, United Kingdom	33
london.leggi_total_mt	28.61	27.9–30.0	MtCO2e/yr	LEGGI 2023 (interim, Dec 2025), territorial; 2021-2022 were 29.8-30.0	34
london.consumption_based_mt	87.95	80.7–88.0	MtCO2e/yr	London consumption-based emissions 2022 (University of Leeds for GLA); 80.7 in 2021	35
london.consumption_food_mt	9.800		MtCO2e/yr	Food within London's consumption-based emissions, 2022	35
uk.territorial_mt	373.4		MtCO2e/yr	UK territorial GHG emissions 2024 (final)	36
uk.consumption_mt	699.2		MtCO2e/yr	UK consumption footprint 2023	37
uk.agriculture_mt	46.52		MtCO2e/yr	UK agriculture 2024	36
uk.beef_enteric_mt	11.33		MtCO2e/yr	Enteric fermentation, beef cattle, 2024 (AR5)	36
uk.beef_manure_mt	2.680		MtCO2e/yr	Manure management, beef cattle, 2024 (AR5)	36
consumption.beef_cwe_kg_per_cap	15.80	11.2–16.9	kg cwe/person/yr	UK supply 2024: (929 + 307 - 142) kt / 69.3 M. High: FAO food balance 16.85 (2023). Low: consistent with NDNS beef intake 13.3 g/day as eaten	38
consumption.dairy_origin_share	0.5000	0.370–0.550	fraction	About half of UK beef comes from the dairy herd	15
conversion.beef_retail_yield	0.7000	0.620–0.740	kg retail / kg cwe	Fat- and bone-free meat incl. edible offal = 72% of carcass (Poore & Nemecek Table S5); AAFC 0.73; closely trimmed boneless 0.62	14
conversion.beef_cook_yield	0.6900	0.670–0.730	kg cooked / kg raw	Beef mince <12% fat, crumbles, pan-fried 0.69 (0.67-0.70); broiled patty 0.73	39
conversion.household_waste_frac	0.1000	0.0600–0.140	fraction of purchases	10% of red-meat purchases wasted (poultry 14%, cheese 13%, milk 8%); applied to all foods	40
emissions.beef_post_farm_kg_per_kg	3.000	2.00–4.50	kgCO2e/kg retail	Processing, packaging, transport and retail; P&N beef-herd post-farm = 2.89	14
emissions.pn_beef_herd	99.48		kgCO2e/kg retail	Global mean, beef herd (AR5 with feedbacks); median 60.36; 5th-95th 37.6-269	14
emissions.pn_beef_dairy	33.30		kgCO2e/kg retail	Global mean, dairy herd; median 34.14	14
emissions.pn_uk_beef_herd	48.40		kgCO2e/kg retail	Poore & Nemecek UK producers, to retail (quoted by CIEL)	15
emissions.pn_uk_beef_dairy	25.90		kgCO2e/kg retail	Poore & Nemecek UK producers, to retail (quoted by CIEL)	15
emissions.diet_fp_t_per_cap	2.000	1.10–2.60	tCO2e/person/yr	Average UK diet (LCA-based); London input-output estimate is 1.1	41
emissions.coc_beef_kg_per_kg_carcass	144.0		kgCO2e/kg carcass	Carbon opportunity cost of land, beef (4% discount); production adds 44	42
bioavailability.haem_abs	0.2500	0.150–0.350	fraction	Haem iron absorption 25% (EFSA 2015; FAO/WHO 10-40%)	8
bioavailability.nonhaem_abs	0.1000	0.0500–0.150	fraction	Non-haem iron; whole mixed diets 14-18%, vegetarian 5-12%	9
bioavailability.zinc_abs_animal	0.3500	0.300–0.400	fraction	Low-phytate diets: FAO/WHO 50/30/15% at phytate:zinc <5/5-15/>15; IZiNCG 26-34% mixed diets	10
bioavailability.zinc_abs_mixed	0.2500	0.200–0.300	fraction	Moderate-phytate foods (tofu, fortified foods); mycoprotein ileal Zn absorption 24% (Edwards & Cummings 2010)	11
bioavailability.zinc_abs_phytate	0.1800	0.120–0.250	fraction	Unrefined cereal / legume diets: IZiNCG 18% (men), 25% (women)	12
bioavailability.zinc_abs_supplement	0.4000	0.300–0.500	fraction	Zinc salts taken with a meal	11

Parameter	Value	Range	Unit	Note	Ref.
bioavailability.b12_abs_meat	0.6000	0.500–0.700	fraction	Mutton 56-89%, chicken 61-66%	13
bioavailability.b12_abs_fish	0.4200	0.350–0.500	fraction	Trout 42%	13
bioavailability.b12_abs_egg	0.09000	0.0400–0.200	fraction	Egg B12 poorly absorbed (<9%)	13
bioavailability.b12_abs_dairy	0.6500	0.500–0.700	fraction	Milk B12	13
bioavailability.b12_abs_synthetic	0.5000	0.400–0.600	fraction	Cyanocobalamin in fortified foods/supplements at low doses	13
bioavailability.b12_abs_none	0		fraction	No B12	
imperial.students	23,920		headcount	31 Dec 2025	43
imperial.staff_fte	8,941		FTE	31 Dec 2025	43
imperial.total_ghg_t	247,453		tCO2e/yr	2024-25: scope 1 33,614; scope 2 12,056; scope 3 201,783	44
imperial.scope12_t	45,670		tCO2e/yr	2024-25 scope 1 + 2	44
imperial.catering_ghg_t	2,097		tCO2e/yr	2023-24 spend-based: food 729, beverages 38, catering equipment/services 1,330	45
campus.beef_t_per_year	9.500	8.00–10.6	t product/yr	Taste Imperial beef purchases 2022/23 10.5 t, 2023/24 8.5 t (FOI)	6
campus.beef_kg_per_meal	0.1200	0.100–0.140	kg/meal	Standard meat portion before the 2024 cut to 100 g	3
campus.meals_per_year	1,000,000	600,000–1,600,000	main meals/yr	Tens of thousands of customers a day across >30 outlets; main meals only	assumption
campus.frequent_beef_meals_per_week	2.000		meals/week	Illustrative frequent campus beef eater	assumption
campus.switch_concentration	25.00		Dirichlet concentration	Uncertainty around destination shares	assumption
campus.switch.other_meat	0.5000		share	Chicken, pork and wild venison (FOI purchase trends)	6,46
campus.switch.fish	0.06000		share		assumption
campus.switch.vegetarian	0.2000		share	Vegetarian sales rose on no-beef days	46,47
campus.switch.vegan	0.1000		share		assumption
campus.switch.off_campus_beef	0.07000		share	Buy beef elsewhere (behavioural leakage)	46,48
campus.switch.off_campus_other	0.07000		share	Eat elsewhere, not beef	48
welfare.meat_nest_share	0.5500	0.450–0.650	share of meals	Meat dishes incl. beef; Oxford baseline 61%	47,49
welfare.nest_lambda	0.6000	0.300–1.00		Nested-logit dissimilarity of the meat nest	assumption
welfare.dish_elasticity	−1.500	−3.00–−0.800		Own-price elasticity of a beef dish	50,51
welfare.meal_price_gbp	6.000	4.50–7.50	GBP	Campus hot meal	assumption
welfare.outside_option_cost_gbp	3.000	1.50–5.00	GBP	Time and price premium of buying a beef meal off campus	assumption
carbon_value.desnz_central_gbp	277.2	139–416	GBP(2022)/tCO2e	2026 central, low and high series	52
carbon_value.epa_scc_usd	215.0		USD(2020)/tCO2	2026, 2% discount rate	53
evaluation.zinc_sd_mg	3.500		mg/day	SD of usual zinc intake, young adults (approx.)	54
evaluation.detectable_zinc_mg	0.5000		mg/day	Minimum detectable difference	assumption
evaluation.alpha	0.05000			Two-sided	
evaluation.power	0.8000				

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